

Enhancing Real-Time Analytics: Streaming Data Quality Metrics for Continuous Monitoring

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ABSTRACT

In today's data-driven landscape, the quality of streaming data is crucial for ensuring the reliability and effectiveness of real-time analytics and decision-making. This article proposes data quality metrics designed specifically for monitoring and assessing data quality in streaming data block systems. The metrics can incorporate data quality dimensions such as timeliness, accuracy, completeness, and consistency to provide a comprehensive evaluation of data quality in dynamic and continuous data streams. The metrics are presented and discussed, along with a case study demonstrating its practical application and impact.

CCS CONCEPTS

• Information systems → Data analytics; Online analytical processing; Data streaming; • Mathematics of computing → Exploratory data analysis; Statistical graphics.

KEYWORDS

Data Quality Metrics, Monitor, Alert, Data Streaming

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1 INTRODUCTION

The role of data extends beyond disciplinary boundaries, representing a foundational element across diverse domains and contexts. Data acts as an important asset for innovation and progress, evident in many fields such as healthcare, finance, education, agriculture, and beyond. The analytical insights extracted from data enable informed decision-making, facilitating transformative change and evidence-based strategies. The importance of data underscores its indispensable role in fostering adaptation, evolution, and prosperity

within our increasingly data-centric society, profoundly shaping the trajectory of our collective advancement.

The importance of data quality is therefore critical. Data quality dimensions such reliability, accuracy, and completeness of data directly impact its usefulness and effectiveness. High-quality data ensures that insights and decisions derived from analysis are reliable and trustworthy. Therefore, while data is abundant, ensuring its quality is critical for maximizing its potential impact and ensuring that data-driven decisions lead to positive outcomes.

In streaming data scenarios, the importance of data quality becomes even more evident due to the real-time and continuous nature of data influx. In these contexts, data is generated rapidly and continuously from various sources (e.g., sensors, IoT devices). The quality of this streaming data directly influences the reliability and timeliness of the insights and actions that can be derived from it. Data quality issues such as inaccuracies, inconsistencies, or missing values can lead to erroneous insights and decisions, potentially causing operational disruptions or missed opportunities. Therefore, in streaming data scenarios, ensuring high-quality data is imperative to leverage the full potential of real-time data analytics, drive actionable insights, and maintain competitive advantage. Therefore, monitoring data quality throughout its lifecycle is essential in streaming data scenarios. Continuous monitoring allows organizations to detect and address data quality issues promptly, ensuring the reliability and integrity of real-time insights and decision-making. Implementing automated validation helps to identify anomalies or deviations from expected data patterns. For example, validating sensor data against predefined thresholds can detect outliers or errors in real-time, enabling immediate corrective actions. Defining and tracking data quality metrics and key performance indicators allows organizations to quantitatively assess data quality over time. Moreover, setting up automated alerts for data quality issues enables proactive identification and response to potential problems.

The current paper presents data quality metrics designed for streaming data scenarios, in particular for growing datasets (i.e., data is continuously generated that should be integrated with previous data), aiming to assess and maintain the reliability and effectiveness of real-time data analytics. These metrics are applied within a system designed for monitoring streamed data blocks, enabling continuous data quality monitoring and real-time alerting capabilities, as presented in [13].

The remaining structure of this article is as follows. Section 2 presents the literature review, examining existing research and

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methodologies related to data quality metrics in streaming data scenarios. Section 3 delves into the details of the data quality metrics. In Section 4, a case study is presented to demonstrate the practical application and utility of the metrics. Finally, Section 5 consolidates the findings and conclusions, providing insights into future directions for research and development in the area of data quality monitoring and alerting for streamed data blocks.

2 LITERATURE REVIEW

Enhancing data quality is crucial as it addresses various issues that result in noisy data, potentially leading to inaccurate or insufficient analysis.

Data quality dimensions encompass specific attributes and characteristics used to evaluate and gauge the overall quality and reliability of data. These dimensions serve as fundamental criteria for assessing data accuracy, completeness, consistency, and other essential facets that determine the value and usability of data within an organization.

In [2], the authors introduced a comprehensive classification framework centered on key quality dimensions: Accuracy, Completeness, Redundancy, Readability, Accessibility, Consistency, Usefulness, and Trust. They detailed various metrics associated with each dimension, providing measure values for evaluation. Accuracy refers to the correctness and precision of the data. Completeness measures whether all required data components are present. Redundancy assesses the presence of duplicated or unnecessary data within a dataset. Readability focuses on how easily data can be interpreted and understood. Accessibility refers to the ease with which data can be accessed by authorized users. Consistency examines the uniformity and standardization of data across the dataset. Usefulness evaluates whether data can effectively support its intended purposes. And, trust measures the reliability and credibility of data. [2] is one of the many approaches to classify data quality issues and in [11], the authors, besides presenting a comprehensive examination of the detrimental impacts associated with poor data quality, presenting an extensive array of data quality dimensions and corresponding measurement methodologies, provide a thorough comparison of proposed methodologies and frameworks documented in the literature, analysing several approaches used to data quality assessment and management. In [10], quality dimensions are categorized into three groups: Intrinsic, Contextual, and Qualitative. The first two categories, focus on a set of quality dimensions. Intrinsic dimensions emphasize the inherent characteristics of the data, while Contextual dimensions consider the environment in which the data is gathered, stored, and used. Qualitative dimensions focus on subjective aspects of data quality, like perceptions and interpretations. Although literature suggests various dimensions, practical implementations prioritise those that can be effectively measured and applied across the organization [12], ensuring a uniform approach to data quality management. It's worth noting that many data quality dimensions and cleaning techniques are applicable across domains and might not address domain-specific requirements [9].

Also, in [7], the authors conduct a review on existing context-aware data quality assessment solutions, particularly focusing on

the challenges posed by big data. They highlight the inadequacies of traditional data quality technologies in addressing the complexities of large and heterogeneous datasets. While conventional approaches rely on predefined rules within the ETL process to define data context, the authors emphasize the need for adaptive and context-sensitive quality assessment methods to cope with the dynamic nature of big data. The review reveals that current solutions often offer limited context awareness, addressing only specific facets of the overall context. Through a comparative analysis of existing models and solutions, the authors advocate for a comprehensive approach to context-aware data quality assessment, culminating in a methodological framework that outlines recommendations for designing and implementing effective context-aware data quality services tailored for big data environments.

While metrics and measures are more frequently used, the availability of a score or index to assess and summarize the overall quality of a dataset based on predefined data quality dimensions is less widespread. In [1], the authors analyze 15 data quality indexes (DQIs) and highlight significant limitations among existing ones, including insufficient or inaccurate information about the used metrics. Additionally, the remarks that certain DQIs are overly context-specific and challenging for non-experts to interpret. Building upon this research, the authors propose a new DQI based on the framework introduced by [14], enhanced with general rule-based metrics designed for context-dependent data quality dimensions. The authors acknowledge certain limitations and areas for improvement, conceptualising this study as a starting point for future research.

In [15], a platform to support the automated and incremental validation of data quality is presented. As the authors suggest, the concept is to allow users to create declaratively "unit tests" (i.e., constraints on the data) for the data. Domain areas such as the Internet of Things (IoT) represent a natural environment for the incremental validation of data since (typically) large amounts of data are streamed by several devices in real time. When data quality is not properly handled, it results in the accumulation of dark data [4]. Dark data, characterized by its unstructured nature and lack of analysis, is accumulated by IoT systems for various reasons, such as compliance and security requirements. Valuable data becomes obsolete due to the absence of tools and processes to use it promptly. Dark data often signifies missed opportunities for businesses due to the uncertainty surrounding its content and quality [3]. In IoT scenarios, large amounts of data are classified as dark data [8]. By swiftly processing and providing data quality metrics data can be used before it becomes unexplored and unstructured. This evidence in the literature reinforces the need for a thorough examination of data quality issues and management techniques, specifically those enhancing and sustaining data quality.

In [13] it is introduced a framework designed to monitor data quality and optimize data reliability for actionable business insights. The framework offers a flexible and scalable approach to consume, analyze, and preprocess data for effective decision-making processes. The framework's classification of data quality streamlines issue resolution by identifying common characteristics and solutions, enhancing decision-making accuracy. In the framework, a Data Layer acts as a central data retention system, managing incoming data blocks from diverse sources and facilitating access for processing. The Data Quality Layer (DQL) processes each data block

and collaborates with the Data Quality Analyzer (DQA) to compute the data quality scores. Each data block undergoes rule-based evaluation to generate an average quality ratio, with historical data stored for comparison and trend analysis on the data quality. Additionally, The DQL uses a *t*-Digest data structure [5], enabling the creation of anomaly detectors (see [6]) to identify deviations from what can be considered "normal" for a given input. The *t*-Digest is available as an open-source project¹. The DQA applies predefined rules categorized into dimensions (e.g., consistency, accuracy) to contribute to the overall data quality score. After evaluation, data blocks and their scores are forwarded to specialized services (e.g., alerting, data loading) for further processing.

3 DATA QUALITY SCORES

The Weighted Quality Score (WQS) is a metric that assesses the adherence of data to predefined quality rules, where each rule (r) and dimension (d) are assigned a specific weight. This score provides a snapshot evaluation of the quality of data at a particular block (j) of data, based on weighted criteria.

For defining WQS, let us consider that:

- D is the set of quality dimensions which will be considered,
- R_d is the set of rules of a specific dimension $d \in D$,
- n_j is the number of rows that the current block j of data contains,
- h_j^r is the number of rows that are in accordance (i.e., hits) with rule $r \in R_d$ of dimension d on block j ,
- w_d is the weight of dimension $d \in D$,
- w_r is the weight of rule r of dimension d , i.e. $r \in R_d$.

The WQS of block j , i.e. WQS_j , is computed by summing the weighted ratio of rows that follow the rules (i.e., hits) across all dimensions $d \in D$ and their respective rules $r \in R_d$:

$$WQS_j = \sum_{d \in D} w_d \left(\sum_{r \in R_d} w_r \frac{h_j^r}{n_j} \right) \quad (1)$$

where $w_d \geq 0, \forall d \in D$ and $w_r \geq 0, \forall r \in R_d$, are such that:

$$\sum_{d \in D} w_d = 1 \quad \text{and} \quad \sum_{r \in R_d} w_r = 1. \quad (2)$$

The Longitudinal Weighted Quality Score (LWQS) is a metric that scores the data quality over time while giving more importance or relevance to newer data compared to older data. For this purpose, for a given block j , the weight attributed to the k^{th} block before j , is modeled using an increasing function $f_k \equiv f(k)$. Several functions can be used for this purpose. In Section 4, an example of such a function is presented.

Equation (3) gives the LWQS of block j , i.e. $LWQS_j$. It is analogous to the previously discussed equation (1) but is tailored specifically for historical data analysis. For $j > 1$, the $LWQS_j$ is given as:

$$LWQS_j = \sum_{d \in D} w_d \left(\sum_{r \in R_d} w_r \frac{H_j^r}{N_j} \right), \quad (3)$$

where the weighted number of hits of rule r is given as:

$$H_j^r = \frac{1}{F_k} \sum_{k=1}^{j-1} f_k h_k^r,$$

the accumulated number of rows is given as:

$$N_j = \frac{1}{F_k} \sum_{k=1}^{j-1} f_k n_k,$$

and

$$F_k = \sum_{k=1}^{j-1} f_k.$$

Equation (3) computes a weighted aggregation of historical data elements across various dimensions and rules, incorporating weights to prioritize different aspects of the analysis. The interpretation and application of this equation enable meaningful insights systematically evaluating and combining historical patterns and behaviors within specified dimensions and rules emphasizing the most recent data.

The Quality Score Delta (QSD) metric, which measures the difference between the WQS and the LWQS (see equation 4), offers significant analytical value for monitoring and enhancing data quality over time within organizational contexts. This metric provides information on the dynamic nature of data quality trends, enabling informed decision-making to refine data management practices continuously. Thus, QSD_j , with $j \geq 1$ is defined as:

$$QSD_j = WQS_j - LWQS_j, \quad (4)$$

with $LWQS_1 = 0$.

Employing the QSD metric effectively enables organizations to assess changes in data quality performance across different periods, thereby facilitating proactive management of data quality issues and targeted improvement initiatives. Positive or negative variations in QSD can signal improvements or declines in data quality, helping to identify emerging trends and patterns.

Furthermore, the QSD metric serves as a strategic tool for evaluating the impact of quality enhancement interventions. By comparing QSD before and after implementing improvement measures, organizations can assess the effectiveness of these interventions and refine strategies accordingly. Setting quality improvement goals based on historical benchmarks using the QSD metric allows organizations to set realistic targets for continuous data quality improvement. Comparing the QSD across various business units or projects helps identify areas that require prioritized attention and resources to enhance data quality effectively.

Moreover, the QSD metric supports risk mitigation efforts by identifying potential risks associated with declining data quality trends. This proactive approach enables organizations to address underlying issues preemptively, safeguarding business operations and decision-making processes from adverse impacts of data quality deterioration.

4 CASE STUDY

For the case study presented here, three quality dimensions are considered, $D = \{d_1, d_2, d_3\}$. The first dimension comprises a single

¹<https://github.com/tddunning/t-digest>

rule $R_{d_1} = \{r_1\}$, while the second and third dimensions include two rules each, i.e. $R_{d_2} = \{r_2, r_3\}$ and $R_{d_3} = \{r_4, r_5\}$.

A total of 26 data blocks are considered, with an average of 49 rows and a standard deviation of 32 rows. The minimum and maximum number of rows were six and 100. This indicates heterogeneous data blocks (see also Fig. 2).

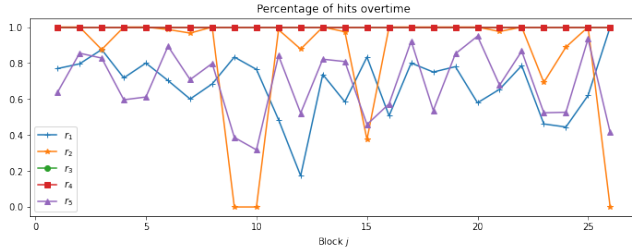


Figure 1: Percentage of hits, for each of the five rules.

From Fig. 1 it is possible to observe a wide diversity of percentage of hits and different behaviors for each rule. Rule r_4 , from dimension d_3 , is always 100%, while the other four rules present percentages from 0 to 100%, with wide variability.

In this study, almost equal weights were attributed to each of the three dimensions. Thus in accordance to equation (2), the weights of each dimension were set to $w_{d_1} = w_{d_2} = 0.33$ and $w_{d_3} = 0.34$, while for the rules the weights were set to $w_{r_1} = 1$ and $w_{r_2} = w_{r_3} = w_{r_4} = w_{r_5} = 0.5$. These weights are used in equations (1) and (3).

For modelling that, more importance or relevance is attributed to newer data when compared to older data, for a block j , the weight attributed to each k^{th} block before j , is given as:

$$f_k \equiv f(k) = \exp\left(-\frac{j-k}{\beta}\right). \quad (5)$$

Here $f(k)$ decreases as k is further away from j and $\beta > 0$ is a parameter that controls the rate of decay. As β increases, the decay rate decreases, and conversely. To illustrate, two values for β were considered, namely $\beta = 2$ and $\beta = 4$. Further, for control purposes, a third case with $f_k = 1$ will be considered. This last case, corresponds to considering that all historical data contributes equitability to LWQS.

Figure 2 presents the results obtained. It is possible to observe that, when considering that all historical data is used, LWQS starts to stabilize, as it corresponds to considering the average percentage of hits. However, when different relevance is given to newer and older ones, the LWQS reacts to changes in the quality of the incoming data. These changes are enhanced when the decay rate, i.e. β , is smaller. In fact, sharper slopes are observed when setting $\beta = 2$, compared to setting $\beta = 4$.

The dimensions of the data blocks, i.e. the number of rows, also influence the LWQS, given its definition (equation (3)). This is also visible in Fig. 2, where the number of rows was multiplied by a factor of $1/200$ to allow visualization.

5 CONCLUSION AND FUTURE WORK

This article underscores the critical importance of data quality and the need for effective measurement tools in the context of growing

datasets. By proposing a data quality score that synthesizes the quality trends across evolving datasets, we contribute to enhancing data management practices and decision-making processes.

However, our work is not without limitations. Future research should focus on refining the data quality score by developing more precise formulas for context-dependent Data Quality Dimensions to better capture the nuances of data quality variations over time and across different data contexts.

In future work, an important direction would be to delve deeper into the weighted nature of the data quality score proposed in this study. This involves refining the weighting scheme used to assign importance to different data quality dimensions within the score calculation. By conducting more extensive research and analysis, we can optimize the weighting methodology to better reflect the relative significance of each dimension in influencing overall data quality. This enhanced understanding of weighted scoring will contribute to more accurate and nuanced data quality assessments, particularly as datasets evolve and grow in complexity. Additionally, exploring dynamic or adaptive weighting approaches that adjust over time based on changing data characteristics and usage patterns could further enhance the utility and relevance of the data quality score in practical applications.

In addition to studying the weighted nature of the data quality score, another promising area for future research is the exploration of automated rule generation through data profiling techniques. By leveraging advanced profiling algorithms and machine learning approaches, we can develop automated methods for identifying and generating data quality rules based on observed data characteristics and patterns. This proactive approach to rule generation will enable the automatic detection and prevention of data quality issues before they impact downstream processes or analyses. Moreover, investigating the integration of profiling-driven rule generation within data quality frameworks can streamline the development and deployment of scalable data quality solutions. This research direction holds significant potential for enhancing the efficiency and effectiveness of data quality management, particularly in dynamic and heterogeneous data environments characterized by large volumes of diverse data sources.

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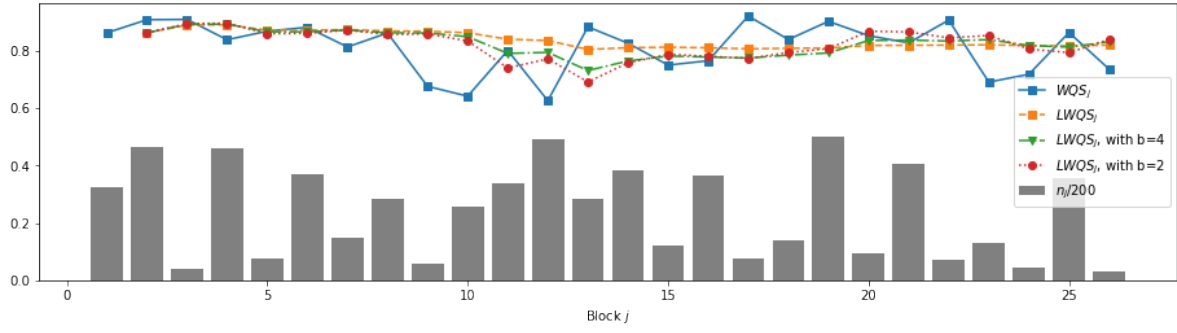


Figure 2: WQS ($\square$), LWQS for $f_k = 1$ ($\square$), $\beta = 4$ (green $\triangle$) and $\beta = 2$ (red $\circ$), and number of rows.

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